In practice, it is crucial to determine under which groups of transformations representations should be considered equivalent, as equivalent representations should be indistinguishable for the chosen similarity measure. Conversely, representations that are not equivalent have to be distinguishable for a similarity measure. In formal terms, this means that a measure has to be *invariant* to exactly those groups of transformations that the underlying representations are equivalent under. We call a representational similarity measure m invariant to a group of transformations $\mathcal{T}$, if for all $\boldsymbol{R} \in \mathbb{R}^{N \times D}$, $\boldsymbol{R'} \in \mathbb{R}^{N \times D'}$ and all $\varphi \in \mathcal{T}(N, D)$, $\varphi' \in \mathcal{T}(N, D')$ it holds that $m(\boldsymbol{R}, \boldsymbol{R'}) = m(\varphi(\boldsymbol{R}), \varphi'(\boldsymbol{R'}))$. Thus, if a measure m is invariant to $\mathcal{T}$, it directly follows that $m(\boldsymbol{R}, \boldsymbol{R}) = m(\boldsymbol{R}, \boldsymbol{R'})$ if $\boldsymbol{R} \sim_{\mathcal{T}} \boldsymbol{R'}$. This implies that a measure can only distinguish representations that are not equivalent under the groups of transformations it is invariant to. Using this notion of invariance, and assuming that representations have identical dimensionality, we can also analyze whether measures satisfy the criteria of a distance metric—in the context of representational similarity, these criteria are typically relaxed to only require $m(\boldsymbol{R}, \boldsymbol{R'}) = 0$ if and only if $\boldsymbol{R} \sim_{\mathcal{T}} \boldsymbol{R'}$ for a group of transformations $\mathcal{T}$ that m is invariant to [155, Apx. A.2].

In the literature [73, 155, 114, 81], there are six main groups of transformations under which representations are considered equivalent, and that representational similarity measures are often designed to be invariant to:

- **Permutations (PT).** A similarity measure m is invariant to permutations if swapping columns of the representation matrices $\boldsymbol{R}$, that is, reordering neurons, does not affect the resulting similarity score. Letting S_D denote the set of all permutations on $\{1, \ldots, D\}$, and for $\pi \in S_D$, $\boldsymbol{P}_\pi = (p_{i,j}) \in \mathbb{R}^{D \times D}$ denote the permutation matrix where $p_{i,j} = 1$ if $\pi(i) = j$ and $p_{i,j} = 0$ otherwise, the group of all permutation transformations is given by
$$\mathcal{T}_{\text{PT}} = \{\boldsymbol{R} \mapsto \boldsymbol{R}\boldsymbol{P}_\pi : \pi \in S_D\}. \tag{3}$$
Permutations neither affect Euclidean distances nor angles between instance representations.
- **Orthogonal Transformations (OT).** As noted in an earlier example, one might intuitively consider two representations equivalent if they can be rotated onto each other. Next to rotations, the group of orthogonal transformations also includes permutations and reflections. Letting $\mathrm{O}(D) := \{\boldsymbol{Q} \in \mathbb{R}^{D \times D}, \boldsymbol{Q}^\mathsf{T}\boldsymbol{Q} = \boldsymbol{I}_D\}$ denote the orthogonal group, the set of these transformations is given by
$$\mathcal{T}_{\text{OT}} = \{\boldsymbol{R} \mapsto \boldsymbol{R}\boldsymbol{Q} : \boldsymbol{Q} \in \mathrm{O}(D)\}. \tag{4}$$
These transformations preserve both Euclidean distances and angles between instance representations.
- **Isotropic Scaling (IS).** Scaling all elements of a representation $\boldsymbol{R}$ identically (isotropic scaling) does not change the angles between instance representations $\boldsymbol{R}_i$. The set of all isotropic scaling transformations is defined as
$$\mathcal{T}_{\text{IS}} = \{\boldsymbol{R} \mapsto a \cdot \boldsymbol{R} : a \in \mathbb{R}_+\}. \tag{5}$$
Isotropic scaling of representations will also rescale the Euclidean distance between instance representations by the same scaling factor a.
- **Invertible Linear Transformations (ILT).** The group of invertible linear transformations, which is defined as
$$\mathcal{T}_{\text{ILT}} = \{\boldsymbol{R} \mapsto \boldsymbol{R}\boldsymbol{A} : \boldsymbol{A} \in \mathrm{GL}(D, \mathbb{R})\}, \tag{6}$$
with $\mathrm{GL}(D, \mathbb{R})$ denoting the general linear group of all invertible matrices $\boldsymbol{A} \in \mathbb{R}^{D \times D}$, forms a broader group of transformations. It includes both orthogonal transformations and rescalings. Both angles and Euclidean distances between instance representations are generally not preserved.
- **Translations (TR).** If the angles between instance representations $\boldsymbol{R}_i$ are not of concern, one might argue that two representations are equivalent if they can be mapped onto each other by adding a constant vector. In that regard, a measure m is invariant to translations if is invariant to the set of all mappings
$$\mathcal{T}_{\text{TR}} = \{\boldsymbol{R} \mapsto \boldsymbol{R} + \boldsymbol{1}_N \boldsymbol{b}^\mathsf{T} : \boldsymbol{b} \in \mathbb{R}^D\}, \tag{7}$$
where $\boldsymbol{1}_N$ is a vector of N ones. Translations preserve Euclidean distances between instance representations.
- **Affine Transformations (AT).** The most general group of transformations that is typically considered for representations is given by the set of affine transformations
$$\mathcal{T}_{\text{AT}} = \{\boldsymbol{R} \mapsto \boldsymbol{R}\boldsymbol{A} + \boldsymbol{1}_N \boldsymbol{b}^\mathsf{T} : \boldsymbol{A} \in \mathrm{GL}(D, \mathbb{R}),\ \boldsymbol{b} \in \mathbb{R}^D\}. \tag{8}$$
This group of transformations in particular also includes rescaling, translations, orthogonal transformations, and invertible linear transformations. Therefore, affine transformations in general do neither preserve angles nor Euclidean distances between instance representations.

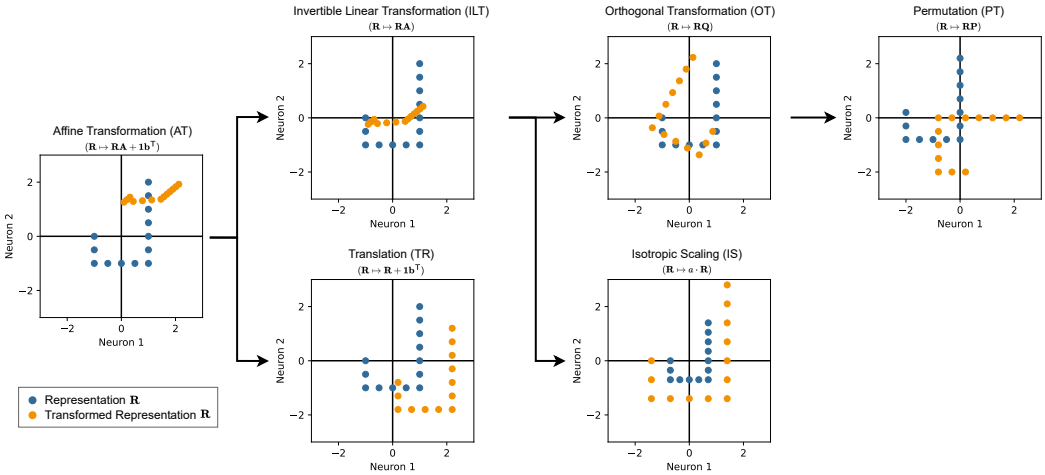

Figure 2: Illustration of representations considered equivalent under different invariances. The invariances form a hierarchy: Arrows describe implication, with the left invariance being more general. For AT and ILT, the same linear transformation is applied. AT further translates representations by the same vector that is used in TR. In OT, the representations are rotated (120°) and reflected over the 15° axis. In PT, axes are swapped. IS applies a scaling factor of 2. See Appendix G for exact parameter values.

We depict the hierarchy of these groups in Figure 2. Table 1 also shows the invariances of all representational similarity measures covered in this survey with respect to these groups. This list of groups of transformations is, however, not exhaustive, and both for practical and theoretical reasons, various other groups may be considered [47]. In practice, neurons are typically assumed to be indexed arbitrarily, so most representational similarity measures are invariant to permutations [73, 65, 81]. Raghu et al. [114] further argued for invariance to invertible linear transformations, as any such transformation could be reverted by a directly following linear layer without altering overall network behavior. However, Kornblith et al. [73] criticized that such invariance leads to unintuitive similarity behavior when $D > N$, such as all representations with full rank being equivalent, and that training of neural networks is not invariant to linear transformations. They argued that orthogonal transformations capture practical differences in representations better.

Preprocessing of Representations. Many representational similarity measures assume certain properties of the representations $\boldsymbol{R}, \boldsymbol{R}'$ that, in practice, are not always given. For instance, it is often assumed that representations are mean-centered in the columns [73, 155, 101], or that they have the same dimensionality. In these cases, the representations need to be preprocessed. There are three kinds of preprocessing that may have to be applied, namely normalization, adjusting dimensionality, and flattening of representations. We discuss these problems in Appendix B.

2.2 Functional Similarity Measures

Functional similarity measures compare neural networks by measuring similarity of their output behavior [30]. Given a set of inputs $\boldsymbol{X}$ and a neural network f that is trained for a classification task on C classes, we let

$$\boldsymbol{O} := f(\boldsymbol{X}) \in \mathbb{R}^{N \times C} \tag{9}$$

denote the matrix of its outputs. Each row $\boldsymbol{O}_i = f(\boldsymbol{X}_i) \in \mathbb{R}^C$ corresponds to the output for input $\boldsymbol{X}_i$. In the context of this survey, we assume that this vector-based output corresponds to *soft predictions*, where each element $\boldsymbol{O}_{i,c}$ denotes the probabilities or decision scores of class c for input $\boldsymbol{X}_i$. From these soft predictions, we can compute the *hard predictions* for a given multiclass classification task via $\hat{c} = \arg\max_c \boldsymbol{O}_{i,c}$, where $\hat{c}$ denotes the predicted class for input $\boldsymbol{X}_i$.

Then, similar to representational similarity measures, functional similarity measures are defined as mappings $m : \mathbb{R}^{N \times C} \times \mathbb{R}^{N \times C} \longrightarrow \mathbb{R}$ that assign a similarity score $m(\boldsymbol{O}, \boldsymbol{O}')$ to a pair of outputs $\boldsymbol{O}, \boldsymbol{O}'$, which are derived from the same inputs $\boldsymbol{X}$. For the compared outputs $\boldsymbol{O}, \boldsymbol{O}' \in \mathbb{R}^{N \times C}$, it is assumed that they are aligned in the sense that the columns $\boldsymbol{O}_{-,c}, \boldsymbol{O}'_{-,c}$ correspond to probability/decision scores of the same class c.

Due to this alignment and the fixed semantics of outputs, analyzing functional similarity generally does not require consideration of preprocessing or invariances. For the same reason, representational similarity measures are unsuitable

for comparison of outputs—the previous assumptions do not hold for representations. Thus, for instance, all permutation invariant measures would consider two outputs that assign 100% probability to different classes equivalent.

Moreover, many functional similarity measures require only black-box access to a model, relying solely on knowledge about inputs and outputs. However, functional similarity measures may include additional information aside from the raw outputs $\boldsymbol{O}_i$. For instance, a set of ground-truth labels $\boldsymbol{y} \in \mathbb{R}^N$ is often given, which is typically used by a quality function q that quantifies how well the output matches the ground-truth. Another kind of additional information are task-based gradients, which, however, require white-box access to the model. Finally, in the context of functional similarity, it is more common that measures compare multiple models at once without relying on pairwise comparisons.

2.3 Relationship Between Representational and Functional Similarity

The notions of representational and functional similarity complement each other (see Fig. 1), and applying both representational and functional similarity measures allows for a more holistic view of neural network similarity [e.g., 137, 69, 52]. To properly interpret potentially conflicting similarity scores stemming from these two perspectives, it is crucial to understand their relationship.

When functional similarity measures indicate dissimilarity, representations must be dissimilar at some layer, assuming that differences in the final classification layer cannot fully explain the functional difference. The opposite is not true: two functionally similar models may use dissimilar representations. Even more, if a functional similarity measure indicates high similarity on a given input set, this does not imply that the compared models are functionally similar in general: high similarity may be the due to easy-to-classify inputs, and out-of-distribution inputs, which tend to amplify functional differences, could yield lower similarity in the corresponding outputs. Similarly, a representational measure indicating high similarity might not generally indicate high functional or representational similarity between models either, as the invariance of a measure might not fit to the given representations.

In conclusion, one generally cannot expect functional and representational measures to correlate, and their scores require contextualization. Only if there is significant functional dissimilarity between two models, there also should be a representational measure indicating significant dissimilarity. Since functional outputs and their similarity measures have a clear and intuitive semantic, this relation can also be used to validate representational similarity measures [34].

3 Representational Similarity Measures

We now review existing representational similarity measures, categorized by their underlying approach to measuring similarity. The categories are illustrated in Figure 3. An overview of all reviewed representational similarity measures can be found in Table 1.

3.1 Canonical Correlation Analysis-Based Measures

Canonical Correlation Analysis (CCA) [60] is a classical method to compare two sets of values of random variables. CCA finds weights $\boldsymbol{w_R} \in \mathbb{R}^D, \boldsymbol{w_{R'}} \in \mathbb{R}^{D'}$ for the columns in the representations, such that the linear combinations $\boldsymbol{R}\boldsymbol{w_R}$ and $\boldsymbol{R'}\boldsymbol{w_{R'}} \in \mathbb{R}^N$ have maximal correlation. Geometrically, the vectors $\boldsymbol{w_R}, \boldsymbol{w_{R'}}$ are projected to the unit ball in $\mathbb{R}^N$ via their representation matrices, such that their angle is minimal. Assuming mean-centered representations, the first *canonical correlation* ρ is defined as

$$\rho := \rho(\boldsymbol{R}, \boldsymbol{R'}) := \max_{\boldsymbol{w_R}, \boldsymbol{w_{R'}}} \frac{\langle \boldsymbol{R}\boldsymbol{w_R}, \boldsymbol{R'}\boldsymbol{w_{R'}} \rangle}{\|\boldsymbol{R}\boldsymbol{w_R}\| \cdot \|\boldsymbol{R'}\boldsymbol{w_{R'}}\|}. \tag{10}$$

One can find additional canonical correlations ρ_i, that are uncorrelated and thus orthogonally projected to the previous ones. This yields a system of D canonical correlations ρ_i defined as

$$\rho_i := \max_{\boldsymbol{w}_{\boldsymbol{R}}^{(i)}, \boldsymbol{w}_{\boldsymbol{R'}}^{(i)}} \frac{\langle \boldsymbol{R}\boldsymbol{w}_{\boldsymbol{R}}^{(i)}, \boldsymbol{R'}\boldsymbol{w}_{\boldsymbol{R'}}^{(i)} \rangle}{\|\boldsymbol{R}\boldsymbol{w}_{\boldsymbol{R}}^{(i)}\| \cdot \|\boldsymbol{R'}\boldsymbol{w}_{\boldsymbol{R'}}^{(i)}\|} \quad \text{s.t. } \boldsymbol{R}\boldsymbol{w}_{\boldsymbol{R}}^{(j)} \perp \boldsymbol{R}\boldsymbol{w}_{\boldsymbol{R}}^{(i)}, \;\; \boldsymbol{R'}\boldsymbol{w}_{\boldsymbol{R'}}^{(j)} \perp \boldsymbol{R'}\boldsymbol{w}_{\boldsymbol{R'}}^{(i)} \;\; \forall j < i, \tag{11}$$

where $\perp$ means orthogonality. If the representations are (nearly) collinear, regularized *Ridge CCA* [147] can be used.

A single similarity score $m(\boldsymbol{R}, \boldsymbol{R'})$ is then computed by aggregating the canonical correlations ρ_i. Standard aggregation choices used to quantify neural network similarity are the mean canonical correlation m_{CCA} [114, 73, 57] and the mean squared canonical correlation m_{CCA^2} [73, 57], also called *Yanai's generalized coefficient of determination* [159]:

$$m_{\text{CCA}}(\boldsymbol{R}, \boldsymbol{R'}) = \frac{1}{D} \sum_{i=1}^{D} \rho_i, \qquad m_{\text{CCA}^2}(\boldsymbol{R}, \boldsymbol{R'}) = \frac{1}{D} \sum_{i=1}^{D} \rho_i^2. \tag{12}$$